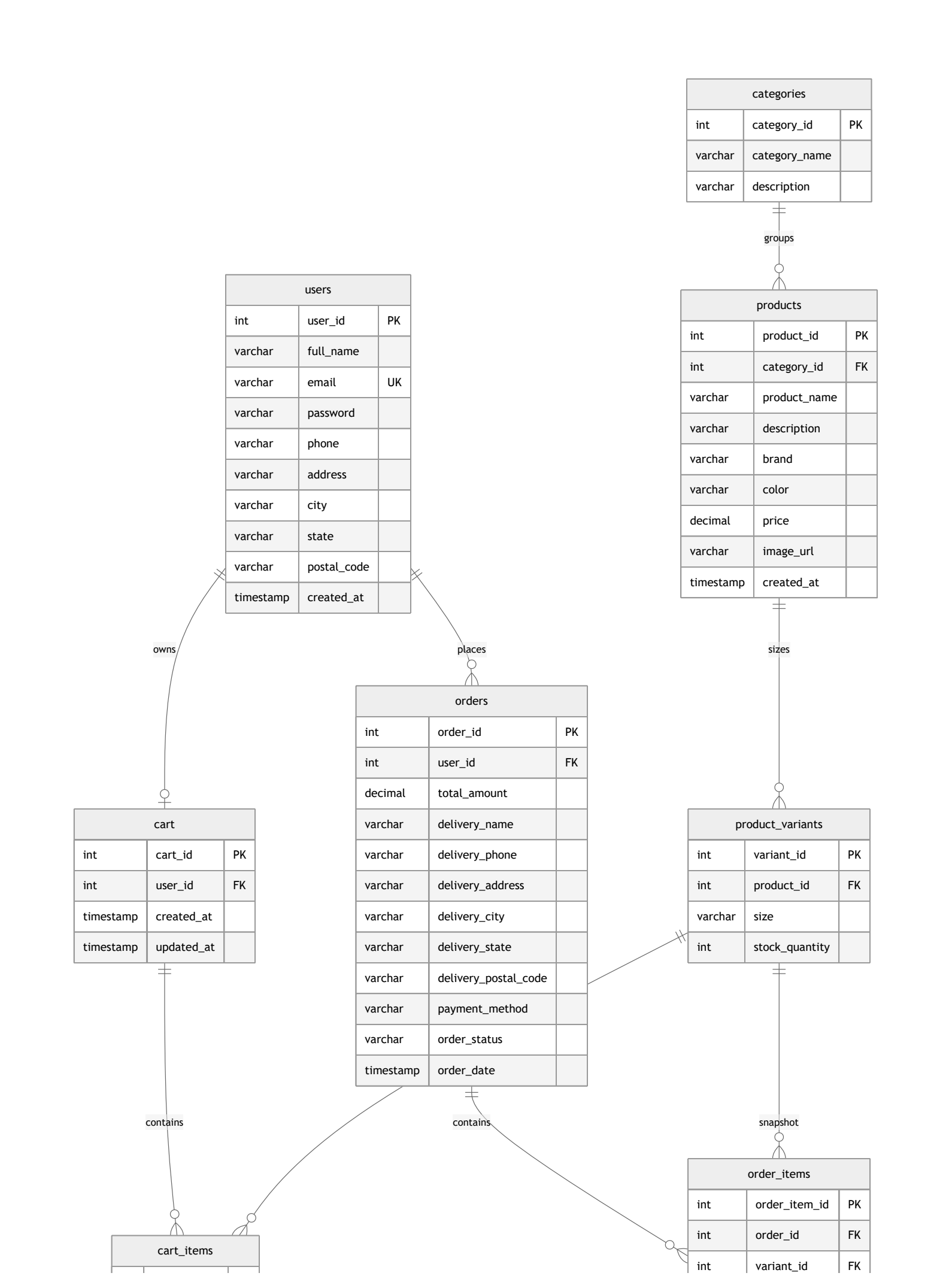
Database

MySQL schema **`fashion_store`** (JDBC URL in `DBConnection`: `jdbc:mysql://localhost:3306/fashion_store`). Access is raw JDBC via `PreparedStatement`; no JPA/Hibernate.

Column lists below come from INSERT/SELECT/map methods in `dao.impl`. Types are the Java types the DAOs bind.

ER diagram


categories
int category_id PK
varchar category_name
varchar description
groups
users
int user_id PK
varchar full_name
varchar email UK
varchar password
varchar phone
varchar address
varchar city
varchar state
varchar postal_code
timestamp created_at
products
int product_id PK
int category_id FK
varchar product_name
varchar description
varchar brand
varchar color
decimal price
varchar image_url
timestamp created_at
owns
places
sizes
orders
int order_id PK
int user_id FK
decimal total_amount
varchar delivery_name
varchar delivery_phone
varchar delivery_address
varchar delivery_city
varchar delivery_state
varchar delivery_postal_code
varchar payment_method
varchar order_status
timestamp order_date
cart
int cart_id PK
int user_id FK
timestamp created_at
timestamp updated_at
product_variants
int variant_id PK
int product_id FK
varchar size
int stock_quantity
contains
contains
snapshot
order_items
int order_item_id PK
int order_id FK
int variant_id FK
cart_items

int	cart_item_id	PK
int	cart_id	FK
int	variant_id	FK
int	quantity	

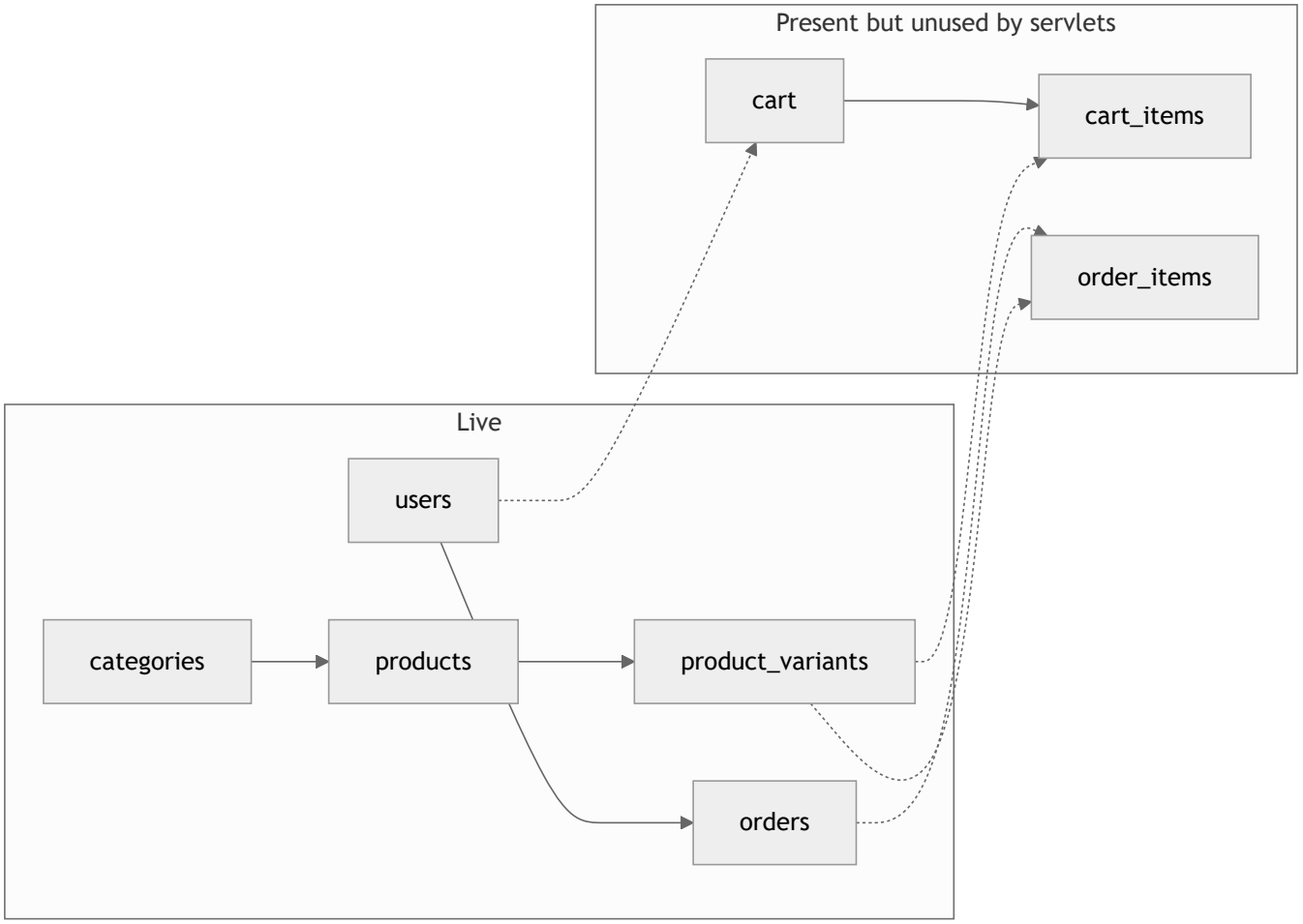
varchar	product_name	
varchar	size	
varchar	color	
int	quantity	
decimal	unit_price	
decimal	subtotal	

Implied uniqueness from DAO lookups: `users.email`, `users.phone`, `cart.user_id` (one cart per user), `product_variants (product_id, size)`.

Table → class → DAO

Table	Model	Interface	Implementation	Used by live UI
`users`	`User`	`UserDAO`	`UserDAOImpl`	Login, Register
`categories`	`Category`	`CategoryDAO`	`CategoryDAOImpl`	Home, Products
`products`	`Product`	`ProductDAO`	`ProductDAOImpl`	Home, Products, Details, Cart
`product_variants`	`ProductVariant`	`ProductVariantDAO`	`ProductVariantDAOImpl`	Details, Cart, Checkout
`cart`	`Cart`	`CartDAO`	`CartDAOImpl`	No (test only)
`cart_items`	`CartItem`	`CartItemDAO`	`CartItemDAOImpl`	No (test only)
`orders`	`Order`	`OrderDAO`	`OrderDAOImpl`	Checkout POST
`order_items`	`OrderItem`	`OrderItemDAO`	`OrderItemDAOImpl`	No (test only)

UI cart lines are `CartItem` objects in **session**, not rows in `cart_items`.



`users` → `User` / `UserDAOImpl`

Column	Java field	Notes
`user_id`	`userId`	Generated key on register
`full_name`	`fullName`	
`email`	`email`	Lookup + uniqueness check
`password`	`password`	Stored as `iterations:salt:hash` (PBKDF2-HMAC-SHA256). Mapper sets `password` to `null` after load
`phone`	`phone`	Optional uniqueness check
`address`	`address`	
`city`	`city`	
`state`	`state`	
`postal_code`	`postalCode`	
`created_at`	`createdAt`	Read only; not in INSERT list (DB default)

DAO writes: `registerUser`, `updateUser`, `updatePassword`. Reads: `loginUser`, `getUserById/Email/Phone`, `getAllUsers`, `emailExists`, `phoneExists`.

`categories` → `Category` / `CategoryDAOImpl`

Column	Java field
`category_id`	`categoryId`
`category_name`	`categoryName`
`description`	`description`

Home and catalog only call `getAllCategories()`. Add/update/delete exist on the DAO but have no admin UI.

`products` → `Product` / `ProductDAOImpl`

Column	Java field
`product_id`	`productId`
`category_id`	`categoryId`
`product_name`	`productName`
`description`	`description`
`brand`	`brand`
`color`	`color`
`price`	`price` (`double` in model; `BigDecimal` in JDBC)
`image_url`	`imageUrl`
`created_at`	`createdAt`

`getAllActiveProducts()` is an alias of `getAllProducts()` — there is no status column.

Size filter joins `product_variants` (`JOIN product_variants v ON p.product_id = v.product_id`).

`product_variants` → `ProductVariant` / `ProductVariantDAOImpl`

Column	Java field
`variant_id`	`variantId`
`product_id`	`productId`

Column	Java field
`size`	`size`
`stock_quantity`	`stockQuantity`

`isStockAvailable` compares `stock_quantity >= requiredQuantity`. `updateStock` exists but checkout never calls it.

`cart` → `Cart` / `CartDAOImpl`

Column	Java field
`cart_id`	`cartId`
`user_id`	`userId`
`created_at`	`createdAt`
`updated_at`	`updatedAt`

INSERT only sets `user_id`. One cart per user via `getCartByUserId` / `getOrCreateCartByUserId`.

`cart_items` → `CartItem` / `CartItemDAOImpl`

Persisted columns:

Column	Java field
`cart_item_id`	`cartItemId`
`cart_id`	`cartId`
`variant_id`	`variantId`
`quantity`	`quantity`

Not stored on this table (UI/session only): `productId`, `productName`, `brand`, `imageUrl`, `size`, `price`. Cart total SQL joins variants → products:

```
cart_items → product_variants → products.price
```

`orders` → `Order` / `OrderDAOImpl`

Column	Java field
`order_id`	`orderId`

Column	Java field
`user_id`	`userId`
`total_amount`	`totalAmount`
`delivery_name`	`deliveryName`
`delivery_phone`	`phone` (`getDeliveryPhone` / `setDeliveryPhone` alias)
`delivery_address`	`deliveryAddress`
`delivery_city`	`deliveryCity`
`delivery_state`	`deliveryState`
`delivery_postal_code`	`deliveryPostalCode`
`payment_method`	`paymentMethod`
`order_status`	`orderStatus`
`order_date`	`orderedAt`

`cancelOrder` sets `CANCELLED` unless already `CANCELLED` or `DELIVERED`. Status update/cancel are not wired to a servlet.

Checkout `placeOrder` does **not** write `order_items` and does not decrement stock.

`order_items` → `OrderItem` / `OrderItemDAOImpl`

Column	Java field
`order_item_id`	`orderItemId`
`order_id`	`orderId`
`variant_id`	`variantId`
`product_name`	`productName`
`size`	`size`
`color`	`color`
`quantity`	`quantity`
`unit_price`	`unitPrice`
`subtotal`	`subtotal`

Name/size/color/price are copied onto the row (order snapshot), not joined at read time.

Confirmation UI reads `lastOrderItems` (`List<CartItem>` in session), not this table.

Connectivity



`DBConnection` loads `com.mysql.cj.jdbc.Driver` once and opens a new connection per call. Implementations close connections with try-with-resources. No connection pool.